

Built a credit scoring model on Bondora's full P2P loan dataset (~915M EUR portfolio, 7 European countries) that **outperforms the platform's available rating variable by +8.6 AUC points** and +49.8% in Gini coefficient — validated on a held-out subset of 25,942 loans.

AUC (OURS)
0.76
vs 0.67 baseline

IMPROVEMENT
+8.6 pts
+49.8% Gini

DECILE LIFT
12.2×
D10 vs D1 bad rate

LOANS MODELED
404k
matured only

01 PROBLEM & APPROACH

Bondora — an Estonian P2P lending platform with ~915M EUR in historical loan volume — publicly releases its complete loan-level dataset, including the platform's risk rating variable per borrower. This provides a rare opportunity: **build a credit scoring model and benchmark it against the published rating**, on identical loans.

02 METHODOLOGY HIGHLIGHTS

- ▶ **Target redefinition.** The raw *is_default* column captured temporary delinquency, missing 78k written-off loans. Constructed a custom *bad_loan* target combining Defaulted + Written Off outcomes, and filtered to matured loans only — avoiding censored-data bias from active loans.
- ▶ **Data leakage prevention.** Identified and excluded 17 of 31 columns describing post-disbursement events; caught a subtle leakage case (*nr_of_payments*) during EDA via cross-class mean ratio analysis.
- ▶ **Feature engineering.** Engineered *income_missing* binary flag from *combined_income* nulls — the strongest single discriminator: 53% bad rate when missing vs 33% when declared.
- ▶ **Two-stage modeling.** Logistic Regression as interpretable benchmark (AUC 0.70); XGBoost as production model (AUC 0.75) capturing non-linear interactions.

03 KEY FINDINGS

METRIC	XGBOOST (OURS)	BONDORA RATING	DELTA
AUC	0.7593	0.6731	+0.086
Gini coefficient	0.5185	0.3462	+49.8%
Decile 1 bad rate	6.5%	—	—
Decile 10 bad rate	79.0%	—	—
Validation sample	25,942 loans with rating available		

The XGBoost model dominates the platform's published rating across the operating range of the ROC curve. Decile analysis shows clean monotonic stratification from 6.5% bad rate in the lowest-risk decile to 79.0% in the highest — a 12.2× lift, above the 8× threshold typically considered "excellent" in industry benchmarks.

04 POTENTIAL BUSINESS APPLICATIONS

- ▶ **Support for risk-based pricing.** The continuous probability output enables differentiated interest rates per risk segment, complementing the current discrete 8-tier letter rating.
- ▶ **Portfolio segmentation.** Deciles 1-4 represent ~32k loans with ~22% average bad rate (lower-risk core, potentially attractive for risk-based pricing). Deciles 8-10 represent ~24k loans with ~69% bad rate (higher-risk tail, candidates for tighter underwriting or collateralization).
- ▶ **Application-time feature set.** All 12 features used are available at the moment of application; no post-disbursement variables, reducing leakage risk for production deployment.

“ The 8.6-point AUC improvement and 12× decile lift suggest a meaningful opportunity for risk segmentation and pricing differentiation, contingent on appropriate out-of-time validation and policy calibration.

LIMITATIONS & CAVEATS

Benchmark is against Bondora's published rating variable, not the platform's full internal risk infrastructure. Model performance is reported on a random stratified split; out-of-time validation is recommended before production use. Feature set is limited to public application-time variables; richer credit bureau data would likely narrow the observed gap.